

Subject Description Form

Subject Code	EE546																					
Subject Title	Electric Energy Storage and New Energy Sources for Electric Vehicles																					
Credit Value	3																					
Level	5																					
Pre-requisite/ Co-requisite/ Exclusion	Nil																					
Objectives	1. To acquire a broad knowledge on classical and modern electric energy storage 2. To understand the development of energy storage from technological, environmental, and societal perspectives.																					
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Understand the importance of energy storage as it pertains to environmental concerns, energy sustainability and climate change. b. Understand various underpinning technologies for conventional and modern energy storage including both portable and stationary systems, such as batteries, supercapacitors, compressed air, flow batteries, new fuel, and fuel cells. c. Explain the role of energy storage in new energy in electric vehicles (EV) and discuss how energy storage devices can be optimally integrated for these applications.																					
Subject Synopsis/ Indicative Syllabus	1. <i>Concept of energy storage</i>: History of energy storage, classification of the types of energy storage. 2. <i>Electrochemical storage</i>: Lead-acid and Nickel batteries, Lithium/sodium-based battery, Flow and Redox batteries, Fuel cell, Sustainability considerations for future electrochemical systems. 3. <i>Carbon-hydride</i>: Carbon hydride energy storage system, non-carbon based fuel, cracking, fuel transportation, fuel storage. 4. <i>Mechanical storage</i>: Compressed air energy storage, pumped hydro energy storage, flywheels. 5. <i>Static Energy Storage</i>: Super-capacitor, Magnetic Energy storage. 6. <i>Electrical energy storage parameters</i>: State of Charge, State of Health, cell impedance and electrochemical impedance spectroscopy, cell models 7. <i>Energy Management System</i>: Battery management, Energy management, cell equalization, conditional monitoring. 8. <i>New Energy for vehicles</i>: Solar vehicles, Fuel cell vehicles, hydrogen engine, compressed gas vehicles, power conversion for new energy.																					
Teaching/Learning Methodology	Delivery of the subject is mainly through formal lectures, complemented by tutorials, worked examples and assignment. Self-learning on the part of students is strongly encouraged and extensive use of web resources will be made. <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="3">Intended subject learning outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th></tr><tr><td>1. Lectures</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>2. Tutorials</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>3. Assignment</td><td>✓</td><td>✓</td><td>✓</td></tr></table>			Teaching/Learning Methodology	Intended subject learning outcomes			a	b	c	1. Lectures	✓	✓	✓	2. Tutorials	✓	✓	✓	3. Assignment	✓	✓	✓
Teaching/Learning Methodology	Intended subject learning outcomes																					
	a	b	c																			
1. Lectures	✓	✓	✓																			
2. Tutorials	✓	✓	✓																			
3. Assignment	✓	✓	✓																			

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed		
			a	b	c
	1. Assignment	25%	✓	✓	✓
	2. Test	25%	✓	✓	✓
	3. Examination	50%	✓	✓	✓
	Total	100 %			
The assignment is designed to assess students' understanding of the energy storage principles and whether they can present the study clearly. It may include take-home assignment and/or mini project. The test is designed to assess students' understanding of the topics that they have learnt relative to learning outcomes (a), (b) and (c). The test is usually conducted in the mid-semester to measure students' performance. Examination: questions are designed to assess learning outcomes (a), (b) and (c). Students are required to answer questions that cover all of the learning outcomes.					
Student Study Effort Expected	Class contact:				
	▪ Lecture				30 Hrs.
	▪ Tutorial and presentation				9 Hrs.
	Other student study effort:				
	▪ Mini project or Assignment				27 Hrs.
	▪ Self-study				49 Hrs.
	Total student study effort				115 Hrs.
Reading List and References	1. "Battery Systems Engineering", A John Wiley & Sons, Ltd., Publication, 2013 2. Sheldon S. Williamson, "Energy Management Strategies for Electric and Plug-in Hybrid Electric Vehicles", Springer New York, 2013 3. Gregory L. Plett, "Battery Management Systems", Boston : Artech House 2015 4. Serguei N. Lvov, Introduction to Electrochemical Science and Engineering. Boca Raton: CRC Press, 2015. 5. G. Pistoia and B.Liaw, "Behaviour of Lithium-Ion Batteries in Electric Vehicles: Battery Health, Performance, Safety, and Cost", Green Energy and Technology, 2018. 6. R.Xiong, "Battery Management Algorithm for Electric Vehicles", 1st ed., Kindle Edition, 2020. 7. Nicolae Tudoroiu, Battery Management Systems of Electric and Hybrid Electric Vehicles, Mdpi AG, 2021 8. Junqiu Li, "Modeling and Simulation of Lithium-ion Power Battery Thermal Management (Key Technologies on New Energy Vehicles) Springer, 2022.				